Laboration #3 (Examination)

Show that the robot can estimate it's position with Kalman filter by help of the encoders and by the laser-scanner. The time, estimated position and uncertainty should be printed out and saved to file with 100ms sample time. The student need to:

- Build and program the robot so it moves from A to B according to a velocity profile.
- Save the estimated positions and the uncertainty from the Odometry, Scan Match and Kalman to file for evaluation, se section recorded file in course description.

The examinations will be done in these steps:

1) Supervisor will point at a random place inside the track. That will be the endpoint and a direction:

2) The end position will then be measured manually by student: (by a ruler or something similar)

3) The end position can then be hardcoded in the code or be used as an argument to the program.

4) Compile the code and run the robot

5) Supervisor will then decide if the robot stops close to the decided point and direction.

6) Send in the uncertainty files

Good Luck!